ARCOT SRI MAHAALAKSHMI WOMENS COLLEGE (AUTONOMOUS)

NAAN MUDHALVAN PROJECT

TOPIC: A CRM APPLICATION FOR THE CHAISHOP&CO

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S. NO	TOPIC	PAGE NO
1	Introduction	1
2	Project Overview	2
3	Problem Statement	3
4	Empathy Map Analysis	4
5	Brainstorming Phase	5
6	Customer Journey Map	6
7	CRM System Workflow Steps	7
8	Functional Requirements	8
9	Data Flow Diagram (DFD)	8
10	Technology Stack	9
11	Project Planning	10
12	Product Backlog & Sprint Planning	11
13	Sprint Workflow	12
14	Salesforce Developer Account Creation	13
15	Object Creation	17
16	Fields & Relationships	19
17	Validation Rule Formulas	21
18	Automation Using Flows	25
19	Reports and Dashboards	27
20	Testing, Advantages & Future Scope	29
21	Conclusion	31

ABSTRACT:

Customer Relationship Management (CRM) systems play an important role in improving business operations and maintaining strong relationships with customers. The project titled "A CRM Application for TheChaiShop&Co" focuses on developing a Salesforce-based CRM system to manage customer information, employer records, and enterprise accounts efficiently. The primary objective of this project is to automate business processes, reduce manual work, and improve the accuracy of customer and sales data management.

The system allows sales representatives and managers to store, update, and manage customer details such as contact information, purchase history, and order details in a centralized platform. By using custom objects, fields, and relationships, the application organizes business data in a structured and accessible format. This helps employees quickly retrieve information and provide better customer service.

One of the key features of this CRM application is the automation of order quantity calculation and total bill generation using formula fields. This reduces manual calculation errors and ensures accurate billing for every transaction. Validation rules are also implemented to ensure that important data such as phone numbers, Aadhaar numbers, and PAN details are entered correctly in the system.

The application also uses Salesforce automation tools such as flows and email alerts to improve customer engagement. A scheduled flow is designed to automatically send birthday greeting emails along with special offers to customers on their birthdays. This feature helps businesses build stronger relationships with their customers and improve customer loyalty.

In addition, the CRM system includes reports and dashboards that allow managers to analyze sales performance and customer activities in real time. These analytical tools help management make informed business decisions and monitor the overall performance of the organization.

Overall, the CRM application for TheChaiShop&Co improves operational efficiency, enhances customer interaction, and provides valuable business insights through automation and data analysis. The project demonstrates how Salesforce technology can be used to build an effective and scalable CRM solution for modern business needs.

1.INTRODUCTION

Customer Relationship Management (CRM) is a system used by organizations to manage interactions with customers, store customer information, and improve business relationships. In modern businesses, maintaining accurate customer data and providing timely services are very important for business growth and customer satisfaction. CRM systems help companies organize customer details, track sales activities, automate business processes, and analyze customer behavior.

The project titled "A CRM Application for TheChaiShop&Co" focuses on developing a cloud-based CRM system using the Salesforce platform. The main goal of this project is to create an efficient application that can manage customer accounts, employer details, and enterprise records in a centralized system. By using Salesforce tools such as custom objects, fields, validation rules, and automation flows, the application helps businesses store and process data effectively.

In many small businesses, customer and sales data are often managed manually using notebooks or spreadsheets. This manual method can lead to errors, data loss, and delays in accessing information. The proposed CRM application solves these problems by providing a structured system that automates order management, billing calculations, and customer communication.

The application also includes important features such as automated order quantity calculation, total bill generation using formula fields, and validation rules to ensure accurate data entry. In addition, automated email notifications are sent to customers on their birthdays with special offers, which helps improve customer engagement and loyalty.

Reports and dashboards are another important part of the system. They allow managers to analyze sales data, track customer activities, and make informed business decisions. With the help of these analytical tools, businesses can understand their performance and plan future strategies effectively.

Overall, this project demonstrates how Salesforce technology can be used to build a powerful CRM solution that improves operational efficiency, reduces manual work, and enhances customer relationship management for TheChaiShop&Co.

2. PROJECT OVERVIEW

The CRM Application for TheChaiShop&Co is developed to improve the way customer information, sales records, and business operations are managed. This application is built using the Salesforce platform, which provides a cloud-based environment for storing and managing business data efficiently. The main purpose of this application is to replace manual record keeping with a digital system that can automate important tasks such as customer management, order processing, billing, and communication.

The application helps businesses maintain a centralized database where all customer and sales information can be stored and accessed easily. Instead of using multiple registers or spreadsheets, employees can manage everything within a single CRM system. This improves accuracy, reduces manual errors, and saves time for employees.

The system is mainly used by sales representatives, store managers, administrators, and business analysts. Sales representatives use the application to enter and update customer details, record orders, and track purchases. Managers use the system to monitor sales activities, view reports, and analyze business performance. Administrators manage the system configuration, create objects and fields, and control user access. Business analysts use the reports and dashboards to evaluate customer trends and sales performance.

One of the main features of the CRM application is customer management. The system stores important customer details such as name, phone number, email address, and purchase history. This allows employees to quickly access customer information and provide better service.

Another important feature is order management. The application allows users to record customer orders, including the products purchased and payment mode. This ensures that every transaction is properly recorded and easily tracked.

The CRM system also includes automated billing, where the total bill amount is calculated automatically using formula fields. This eliminates the need for manual calculations and reduces billing errors.

In addition, the system provides email notifications that automatically send special offers or birthday greetings to customers. This feature helps improve customer engagement and build stronger relationships with customers.

Overall, the CRM application for TheChaiShop&Co helps streamline business operations, improve data management, and support better decision-making through automation and real-time data analysis.

3. PROBLEM STATEMENT

Before implementing the CRM application, TheChaiShop&Co faced several challenges in managing customer data, sales transactions, and communication. Most of the business operations were handled manually using notebooks or spreadsheets. This manual approach created many difficulties such as errors in calculations, delays in accessing information, and lack of proper communication with customers.

Sales representatives often had to calculate order quantities and total bill amounts manually. This increased the chances of mistakes in billing and sometimes resulted in incorrect invoices. Managers also found it difficult to monitor sales performance because the data was scattered across different files and registers. As a result, generating reports and analyzing business performance became time-consuming and inefficient.

Another major issue was the lack of proper communication with customers. Since there was no automated system to send notifications or promotional offers, customers were not informed about special discounts or important updates. This reduced customer engagement and loyalty.

In addition, managing large amounts of customer data manually made it difficult to store, organize, and retrieve information when needed. This led to delays in service and sometimes resulted in loss of important business data.

To overcome these challenges, a CRM system was required to automate processes, improve data accuracy, and enhance communication with customers.

3.1 PROBLEM STATEMENT TABLE

User	Trying To Do	Problem Faced	Reason	Result/Feeling
Sales Representative	Manage customer orders and billing	Manual billing and incorrect calculations	Orders and billing were calculated manually	Stress and inefficiency
Store Manager	Monitor sales performance and customer records	Reports are delayed and incomplete	Data stored in multiple files and registers	Confusion and difficulty in decision-making
Customer	Receive offers and updates from the store	No communication or notifications	No automated communication system	Customers feel disconnected and less valued

4. EMPATHY MAP ANALYSIS

The Empathy Map Analysis helps to understand the needs, behavior, and challenges of different users who interact with the CRM application. By analyzing what users think, feel, see, hear, say, and experience, developers can design a system that improves user experience and solves real business problems.

In the CRM Application for TheChaiShop&Co, the main users are Sales Representatives, Managers, Administrators, and Analysts. Each user has different responsibilities and expectations from the system.

4.1. EMPATHY MAP TABLE

User	Think & Feel	Hear	See	Say & Do	Pain	Gain
Sales Representatives	Want to manage orders quickly and avoid mistakes	Instructions from managers about sales targets	Customer orders and product details	Enter customer data and process orders	Manual calculations and billing errors	Faster order processing and accurate billing
Managers	Want clear reports and smooth business operations	Feedback from sales staff and customers	Sales reports, customer activity, and performance	Monitor sales and make decisions	Difficulty tracking data from multiple sources	Easy access to reports and better decision making
Administrators	Need a well-organized and secure system	Technical updates and system requirements	System settings, user accounts, and	Manage system configuration and user access	Managing large amounts of data manually	Efficient data management and system control

			database structure			
Analysts	Want accurate data for analysis	Business performance discussions	Sales trends, customer patterns, and reports	Analyze data and generate insights	Incomplete or unorganized data	Better data insights and business strategy

5. BRAINSTORMING PHASE

The Brainstorming Phase is an important step in project development where different ideas and possible solutions are discussed before selecting the best one. During this phase, the team analyzes various methods that could solve the problems faced in managing customer data, orders, and billing at TheChaiShop&Co.

Several solutions were considered to improve the existing manual system. Each option was evaluated based on factors such as ease of use, automation capability, data security, scalability, and efficiency.

1. MANUAL SYSTEM

The manual system involves maintaining customer records, orders, and billing information in notebooks or registers. This was the traditional method used in many small businesses.

However, this system has several disadvantages. Manual calculations can lead to errors, data can easily be lost or damaged, and retrieving information takes a lot of time. It also becomes difficult to generate reports or analyze business performance.

2. EXCEL-BASED SYSTEM

Another idea was to use spreadsheet software like Microsoft Excel to store customer data and sales records. Excel allows basic data organization and simple calculations using formulas.

Although this method is better than the manual system, it still has limitations. Data management becomes difficult when the business grows, and there is a higher risk of data duplication or accidental deletion. Excel also lacks advanced automation features and secure multi-user access.

3. WEB-BASED CRM SYSTEM

A web-based CRM system can store customer information and sales data online. This allows multiple users to access the system from different locations.

While this solution provides better accessibility and automation, developing a custom web-based CRM from scratch requires significant time, technical expertise, and maintenance effort.

4. SALESFORCE CRM

Salesforce CRM is a cloud-based platform specifically designed for customer relationship management. It provides powerful tools for customer data management, sales tracking, automated workflows, reporting, and secure cloud storage.

Salesforce also allows developers to create custom objects, fields, automation tools, and dashboards without building the system completely from the beginning.

WHY SALESFORCE WAS SELECTED

After evaluating all the options, Salesforce CRM was selected as the best solution for this project. Salesforce offers many advantages such as cloud-based data storage, automation features, secure access control, scalability, and built-in reporting tools.

It allows businesses to manage customers, orders, and billing efficiently in one centralized platform. Salesforce also supports automated email notifications, formula fields for billing calculations, and dashboards for monitoring business performance.

Therefore, Salesforce CRM provides a reliable, scalable, and efficient solution for improving customer management and sales operations at TheChaiShop&Co.

6. CUSTOMER JOURNEY MAP

The Customer Journey Map explains the workflow of the CRM system and how users interact with the application from the beginning to the final business decision-making stage. It shows the step-by-step process followed by employees when using the CRM Application for TheChaiShop&Co.

This workflow helps the organization manage customer data efficiently, process orders accurately, and generate useful reports for business growth.

7. CRM SYSTEM WORKFLOW STEPS

Step	Process	Description
1	Login	The user logs into the CRM system using their username and password. Secure authentication ensures that only authorized users can access the application.
2	Access Application	After login, the user accesses the CRM dashboard where different modules like customer management, orders, and reports are available.
3	Enter Customer Data	Sales representatives enter customer information such as name, phone number, email, and purchase details into the system. This creates a centralized customer database.
4	Create Orders	When a customer places an order, the sales representative records the order details including product name, quantity, and price in the CRM system.
5	Calculate Bill	The system automatically calculates the total bill amount using predefined formula fields. This reduces manual calculation errors and speeds up the billing process.
6	Generate Reports	Managers and analysts generate reports and dashboards to view sales performance, customer trends, and order history.
7	Business Decision Making	Based on the reports and data insights, managers make strategic decisions such as improving marketing strategies, offering discounts, or planning inventory.

8. FUNCTIONAL REQUIREMENTS

Functional Requirements describe the main operations and features that the CRM application must perform. These requirements define how the system should behave and what functions it should provide to users such as sales representatives, managers, administrators, and analysts.

The CRM Application for TheChaiShop&Co is designed to automate customer management, order processing, billing, and reporting. The following are the important functional requirements of the system.

8.1 FUNCTIONAL REQUIREMENTS TABLE

No	Function	Description
1	User Login	The system should allow users to securely log in using a username and password to access the CRM application.
2	Data Entry	The system should allow users to enter and store customer information such as name, contact details, and purchase history.
3	Order Creation	The system should allow sales representatives to create and manage customer orders including product details, quantity, and price.
4	Billing Automation	The system should automatically calculate the total bill amount using predefined formulas to reduce manual calculation errors.
5	Email Notification	The system should send automated emails to customers for special offers, promotions, and important updates.
6	Reports Generation	The system should generate reports and dashboards to help managers and analysts track sales performance and customer activity.

9. DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD) shows how data moves through the CRM system and how it is processed at different stages. It helps in understanding how information flows from the user to the system and how it is stored, validated, processed, and finally used for reporting and decision-making.

In the CRM Application for TheChaiShop&Co, data flows through several important components of the Salesforce platform.

9.1 DATA FLOW STRUCTURE

User → Salesforce UI → Custom Objects → Validation Rules → Flows → Reports → Dashboards

Step	Component	Explanation
1	User	The process starts with the user such as a sales representative, manager, or administrator. The user enters data like customer details, order information, and billing details into the system.
2	Salesforce UI	The Salesforce User Interface (UI) is the front-end environment where users interact with the CRM system. It provides forms, dashboards, and menus that allow users to input and access data easily.
3	Custom Objects	Custom objects are used to store specific business data such as customer records, orders, and billing information. These objects act like database tables where all related data is organized and stored.
4	Validation Rules	Validation rules ensure that the data entered by the user is accurate and follows certain conditions. For example, they check if required fields are filled or if values are entered in the correct format.
5	Flows	Flows are automation tools in Salesforce that perform actions automatically based on certain conditions. They can automate processes like sending emails, updating records, or calculating billing amounts.
6	Reports	Reports collect and display data stored in the system. Managers and analysts can generate reports to analyze sales performance, customer activity, and order history.
7	Dashboards	Dashboards provide visual representations of reports using charts and graphs. They help managers quickly understand business performance and make strategic decisions.

10. TECHNOLOGY STACK

Layer	Technology
Frontend	Salesforce Lightning
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Flows
Analytics	Reports & Dashboards

11. PROJECT PLANNING

Project Planning is an important step in software development that helps organize the tasks, schedule, resources, and goals required to successfully complete a project. It ensures that the development process follows a structured approach so that the CRM application for TheChaiShop&Co can be completed efficiently and on time.

Effective project planning helps the development team understand the project requirements, divide the work into manageable tasks, and track progress throughout the development process. For this project, modern software development practices such as Agile methodology, sprint planning, development phases, and team collaboration were used to manage the workflow.

1. AGILE METHODOLOGY

The **Agile methodology** is a flexible and iterative approach to software development. Instead of completing the entire project at once, the system is developed in small stages called iterations. Each iteration focuses on developing and improving specific features of the CRM system.

Agile helps the team quickly adapt to changes and improve the system based on feedback. In the CRM application project, Agile allows developers to gradually build important modules such as customer management, order management, billing automation, and reporting.

2. SPRINT PLANNING

Sprint planning is a key activity in Agile development. A sprint is a short development cycle, usually lasting one to two weeks, during which a specific set of tasks is completed.

During sprint planning, the team decides which features or tasks should be developed in that sprint. For example, one sprint may focus on creating the customer management module, while another sprint may focus on order creation and billing automation. This approach helps the team stay organized and maintain steady progress.

3. DEVELOPMENT PHASES

The project development process is divided into several phases to ensure systematic progress. These phases include:

Phase	Description
Requirement Analysis	Understanding the business needs and system requirements.
System Design	Designing system architecture, database structure, and workflow diagrams.
Development	Building the CRM application using Salesforce tools and features.
Testing	Checking the system for errors and ensuring that all features work correctly.
Deployment	Launching the CRM application for actual use.
Maintenance	Updating and improving the system after deployment.

4. TEAM COLLABORATION

Team collaboration is essential for successful project development. The team members communicate regularly to discuss progress, share ideas, and solve problems. Collaboration ensures that everyone understands their responsibilities and works towards the same project goals.

In the CRM project, collaboration between developers, administrators, and analysts helps in designing efficient workflows, implementing automation features, and generating useful reports for business analysis.

12. PRODUCT BACKLOG & SPRINT PLANNING TABLE

Sprint	User Story	Priority	Story Points
Sprint 1	As a user, I want to log in securely to access the CRM system	High	5
Sprint 2	As a sales representative, I want to enter and store customer details	High	8
Sprint 3	As a sales representative, I want to create and manage customer orders	High	8

Sprint 4	As a system, I want to automatically calculate billing amounts	Medium	5
Sprint 5	As a system, I want to send automated email notifications to customers	Medium	5
Sprint 6	As a manager, I want to generate sales reports and dashboards	High	8

13. SPRINT WORKFLOW

The Sprint Workflow describes the steps followed during each sprint in Agile development.

Step	Activity	Description
1	Product Backlog Creation	All project requirements and features are listed as user stories.
2	Sprint Planning	The development team selects user stories to complete in the sprint.
3	Task Development	Developers build and implement the selected features.
4	Testing	The system is tested to ensure the features work correctly.
5	Sprint Review	The completed work is reviewed by the team and stakeholders.
6	Sprint Retrospective	The team discusses improvements for the next sprint.

14. SALESFORCE DEVELOPER ACCOUNT CREATION

Creating a **Salesforce Developer Account** is the first step to start building applications on the Salesforce platform. The Developer Edition is a **free environment** that allows developers to create, test, and customize CRM applications without affecting real business data.

Below are the **step-by-step instructions** for creating a Salesforce Developer Account.

Step 1: Open the Salesforce Developer Website

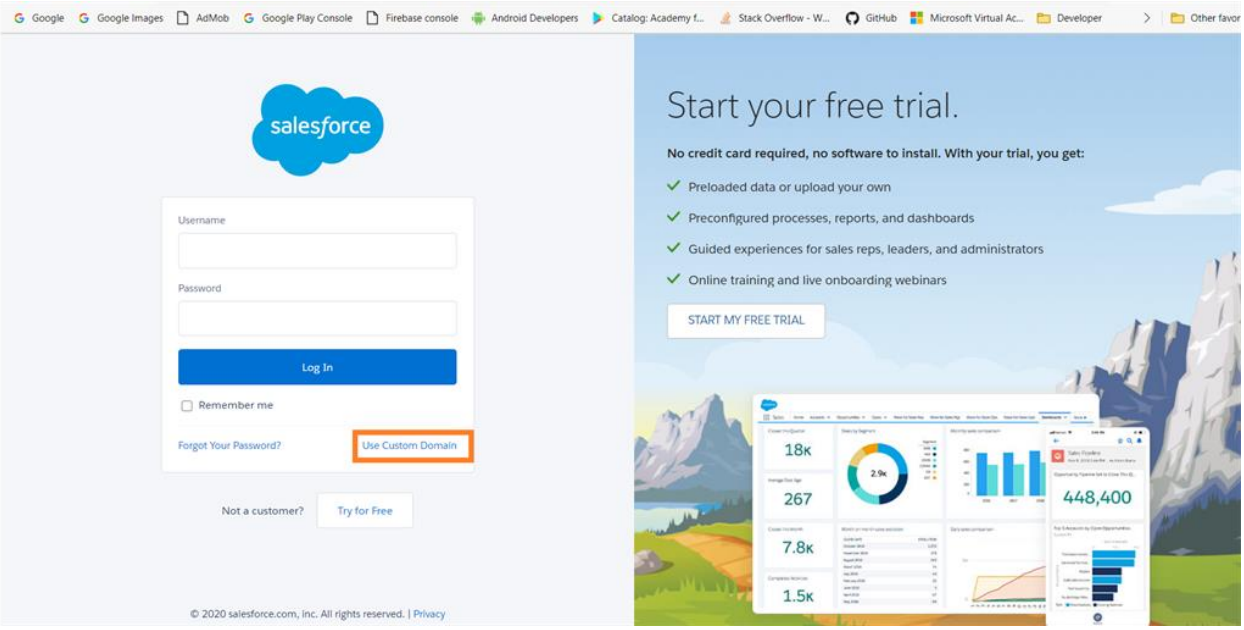


Fig: 14.1

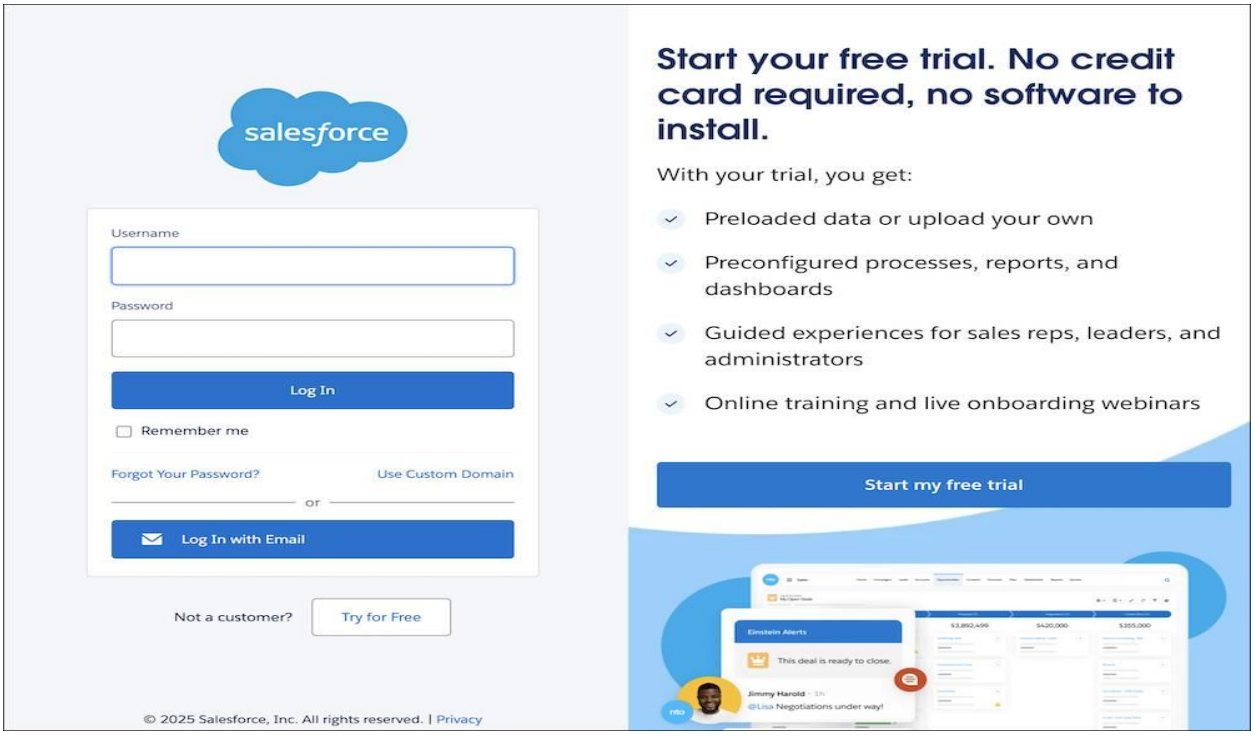


Fig: 14.2

1. Open your web browser.

2. Go to the Salesforce Developer signup page:
   https://developer.salesforce.com/signup
3. Click on the Sign Up or Free Developer Edition option.

This page allows users to register for a free Salesforce development environment.

Step 2: Fill the Signup Form

On the signup page, you must fill in the required details.

Typical fields include:

- First Name
- Last Name
- Email Address
- Company Name
- Role / Job Title
- Country
- Username (must be in email format)

After filling in all details:

1. Accept the Salesforce Terms and Conditions.
2. Click the Sign Me Up button.

The username must be unique and usually written in the form of an email address.

Step 3: Verify Email

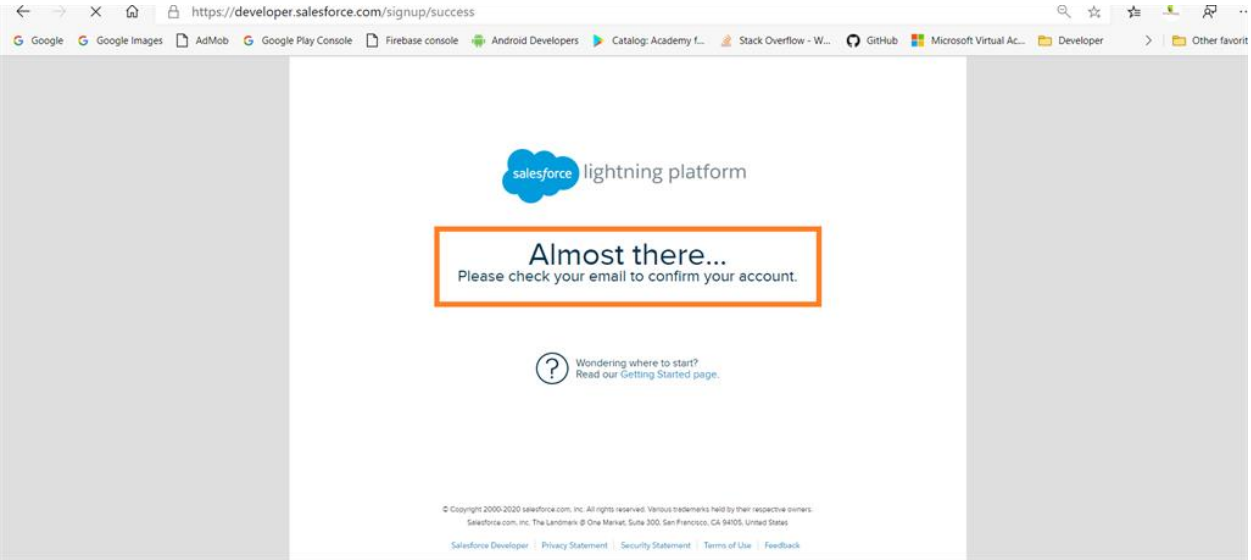


Fig: 14.3

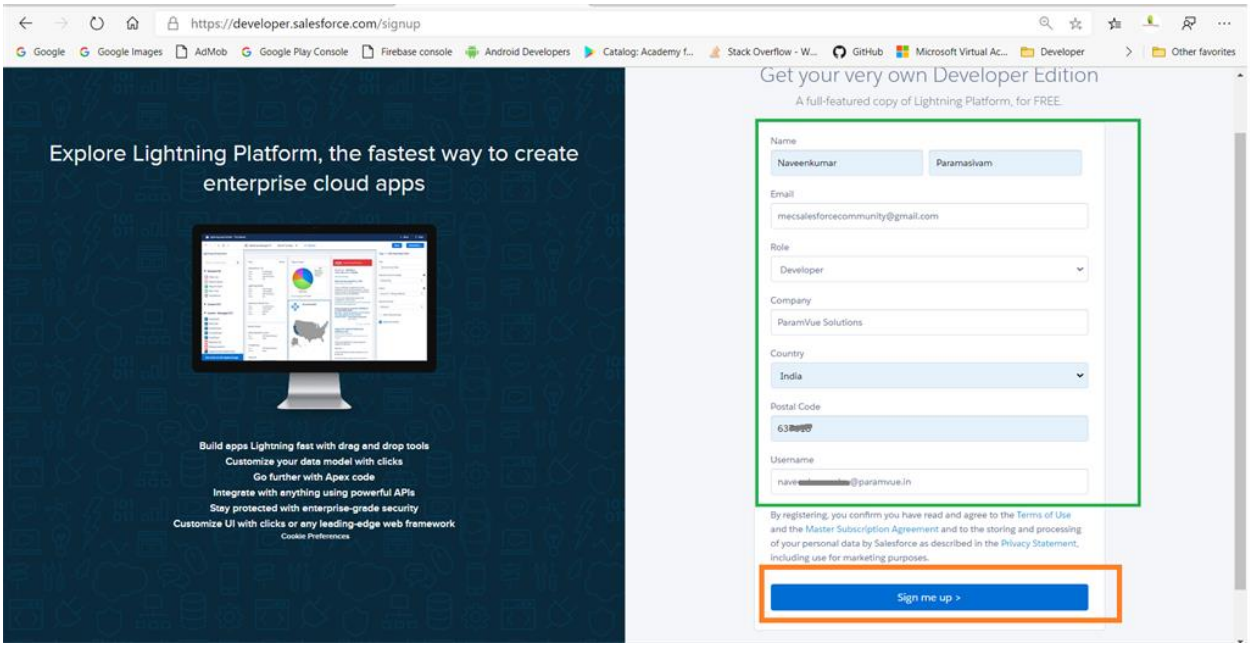


Fig: 14.4

After submitting the form:

1. Salesforce sends a verification email to the registered email address.
2. Open your email inbox.
3. Click the Verify Account link.

This step confirms your email and activates the developer account.

Step 4: Set Password

Once you click the verification link:

1. A **password setup page** will appear.
2. Enter a new password and confirm it.
3. Select a security question and answer for account recovery.
4. Click Change Password.

After completing this step, your Salesforce developer account will be activated.

Step 5: Login to Salesforce

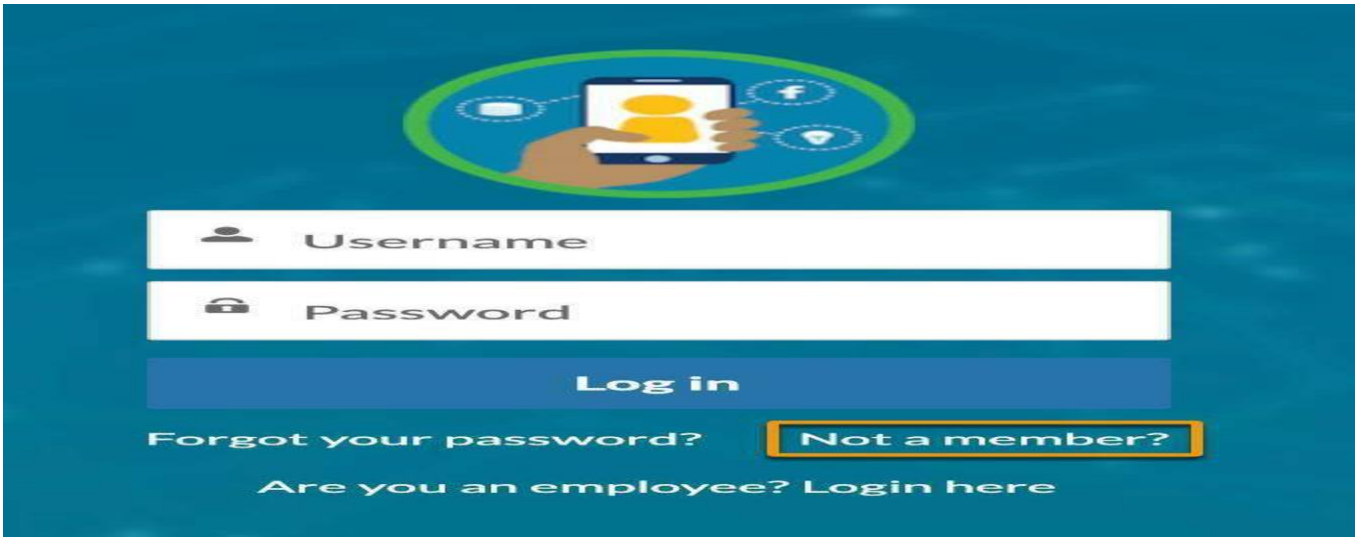


Fig: 14.5

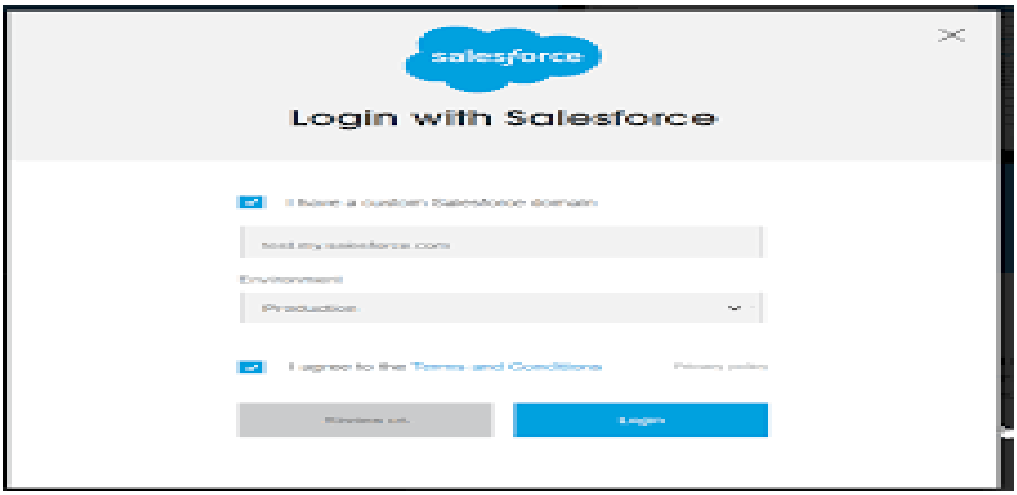


Fig: 14.6

1. Go to the Salesforce login page:
   https://login.salesforce.com
2. Enter your username and password.
3. Click Login.

Once logged in, you will see the Salesforce dashboard, where you can start building CRM applications.

15. OBJECT CREATION

In Salesforce, **objects** are database tables used to store data in an organized way. Each object contains records and fields that hold specific information. Salesforce provides **standard objects** such as Account and Contact, but developers can also create **custom objects** to store business-specific data.

For the CRM Application of TheChaiShop&Co, several **custom objects** were created to manage different types of accounts and store related information. These objects help organize customer data, business accounts, and employer details within the system.

The following custom objects were created in the CRM application:

- Employer Account
- Customer Account
- Enterprise Account

These objects allow the system to store and manage different types of user and business information efficiently.

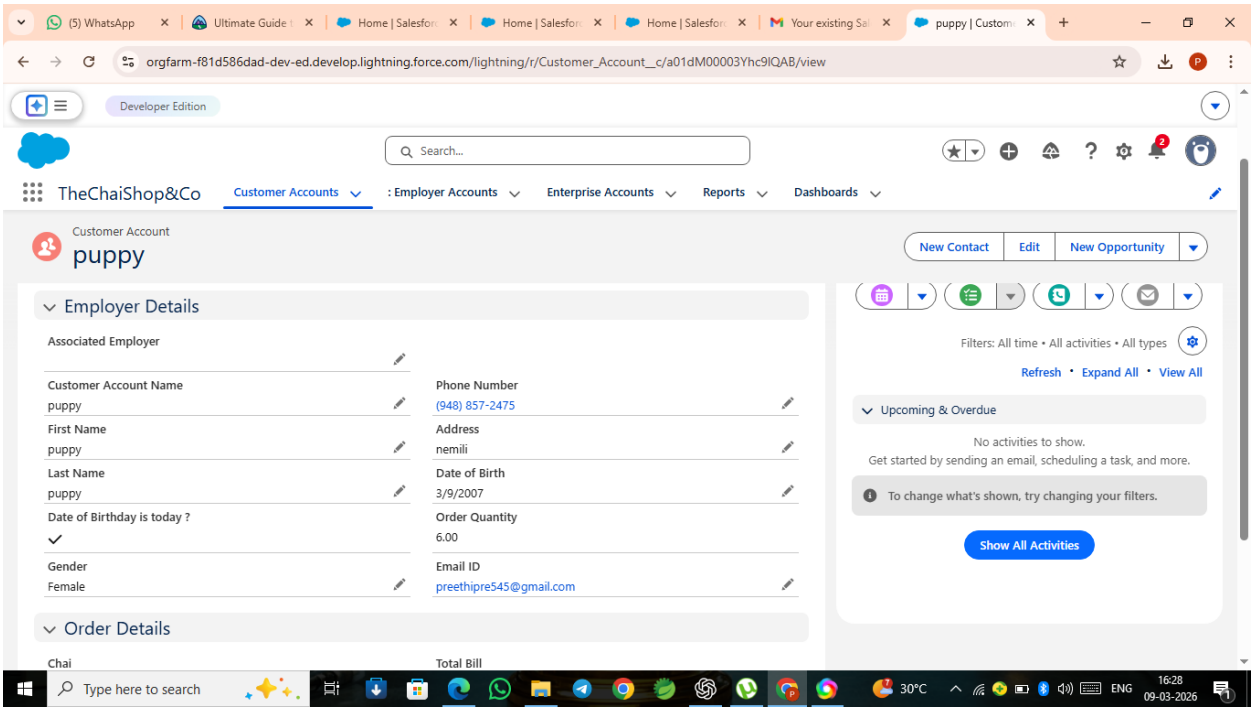


Fig: 15.1

15.1 CUSTOM OBJECTS CREATED

Object Name	Description
Employer Account	Stores details of employers or staff members associated with the business.
Customer Account	Stores customer information such as name, contact details, and purchase history.
Enterprise Account	Stores information about enterprise-level business accounts or partners.

15.2. OBJECT COMPONENTS

1. Labels

Labels represent the display name of the object in Salesforce. The label is what users see in the Salesforce interface when they access the object.

For example:

Object	Label
Employer Account	Employer Account
Customer Account	Customer Account
Enterprise Account	Enterprise Account

Labels help users easily identify objects in the CRM system.

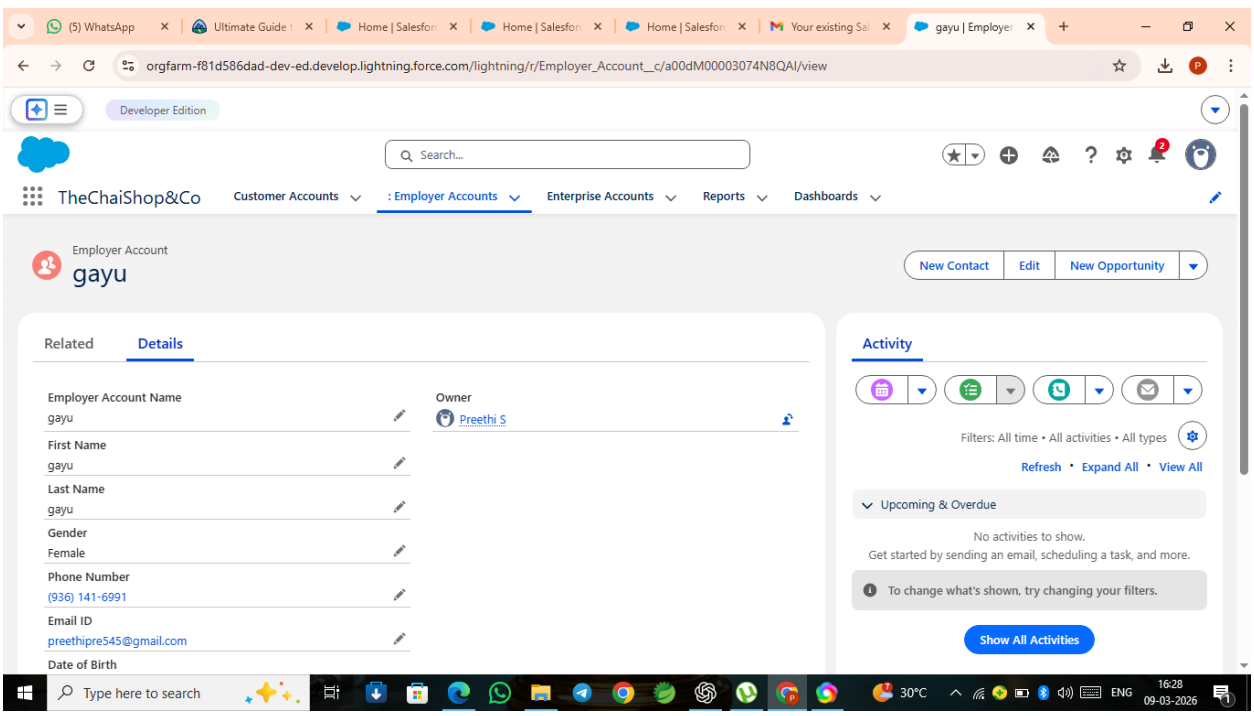


Fig: 15.2

2. Fields

Fields are used to store specific pieces of information within an object. Each field represents a data attribute.

Examples of fields used in the CRM system:

Field Name	Description
Customer Name	Stores the name of the customer
Phone Number	Stores customer contact number
Email	Stores customer email address
Order Amount	Stores the billing amount
Address	Stores customer address

Fields help organize and store detailed information about records in the system.

3. Record Name

The **Record Name** field is the primary identifier for each record in an object. It helps uniquely identify each entry in the system.

For example:

Object	Record Name Field
Employer Account	Employer Name
Customer Account	Customer Name
Enterprise Account	Enterprise Name

Each record name helps users quickly locate and manage records in the CRM application.

16. FIELDS & RELATIONSHIPS

In Salesforce, fields and relationships are used to organize and connect data within objects. Fields store specific information about a record, while relationships connect different objects so that related data can be accessed easily.

In the CRM Application for TheChaiShop&Co, custom fields are created to store detailed customer and account information. Relationships are used to link different objects, making it easier to manage and retrieve related data.

16.1. CUSTOM FIELDS

Custom fields are fields created by developers to store specific business information that is not available in standard fields. These fields help capture important details about customers, employers, and enterprise accounts.

Examples of custom fields used in the CRM system include:

Field Name	Field Type	Description
Name	Text	Stores the name of the customer or account holder
Phone Number	Phone	Stores the contact number of the customer
Email	Email	Stores the customer's email address
Address	Text Area	Stores the residential or business address
Date of Birth	Date	Stores the customer's birth date
Aadhaar Number	Number / Text	Stores the Aadhaar identification number
PAN Number	Text	Stores the PAN card number for identification

These custom fields help maintain complete and organized customer records in the CRM system.

16.2. LOOKUP RELATIONSHIP

A **Lookup Relationship** is a type of connection between two objects in Salesforce. It links one object to another so that related records can be associated with each other.

For example:

- A **Customer Account** can be linked to an **Employer Account**.
- An **Enterprise Account** can be connected to multiple customer records.

This relationship allows the system to display related information across objects. For instance, when viewing a customer record, users can easily see the employer or enterprise account associated with that customer.

16.3. ADVANTAGES OF LOOKUP RELATIONSHIPS

- Connects related records across different objects
- Allows flexible data relationships
- Improves data organization and accessibility

- Helps generate better reports and analysis

17. VALIDATION RULE FORMULAS

Salesforce validation rules use **formula expressions** to define the conditions that must be satisfied.

1. AADHAAR NUMBER VALIDATION FORMULA

This formula checks whether the Aadhaar number contains exactly 12 digits.

LEN(Aadhaar_Number__c) <> 12

If the number entered is not 12 digits long, the system displays an error message.

2. PHONE NUMBER VALIDATION FORMULA

This formula checks whether the phone number contains exactly 10 digits.

LEN(Phone_Number__c) <> 10

If the phone number does not contain 10 digits, the system will prevent the record from being saved.

3. PAN NUMBER VALIDATION FORMULA

This formula ensures that the PAN number follows the correct format.

NOT(REGEX(PAN_Number__c , "[A-Z]{5}[0-9]{4}[A-Z]{1}"))

This expression checks whether the PAN number includes:

- Five uppercase letters
- Four numbers
- One uppercase letter

If the format is incorrect, the system displays an error message.

17.1. AADHAAR NUMBER VALIDATION RULE

Used to check **Aadhaar must be 12 digits and cannot start with 0**.

```
AND(
NOT(ISBLANK(Aadhar_Number__c)),
LEN(Aadhar_Number__c) <> 12,
NOT(REGEX(Aadhar_Number__c,"^[1-9][0-9]{11}$"))
)
```

Error Message

Aadhar number must contain exactly 12 digits and should not start with 0.

17.2. PHONE NUMBER VALIDATION RULE

Ensures **10 digit phone number only**.

```
AND(
NOT(ISBLANK(Phone_Number__c)),
NOT(REGEX(Phone_Number__c,"^[1-9][0-9]{9}$"))
)
```

Error Message

Phone number must contain exactly 10 digits and cannot start with 0.

17.3. PAN NUMBER VALIDATION RULE

Checks valid PAN format.

```
AND(
NOT(ISBLANK(PAN_Details__c)),
NOT(REGEX(PAN_Details__c,"^[A-Z]{5}[0-9]{4}[A-Z]{1}$"))
)
```

Error Message

Enter a valid PAN number format.

17.4. BIRTHDAY CHECK FORMULA FIELD

Checks if **today is customer's birthday**.

```
IF(
AND(
DAY(TODAY()) = DAY(Date_of_Birth__c),
MONTH(TODAY()) = MONTH(Date_of_Birth__c)
),
TRUE,
FALSE
)
```

Return Type → **Checkbox**

17.5. ORDER QUANTITY FORMULA

Counts how many products the customer selected.

```
IF(INCLUDES(Products_Purchased__c,"Platinum Tea Powder"),1,0) +
IF(INCLUDES(Products_Purchased__c,"Double Delight Tea Powder"),1,0) +
IF(INCLUDES(Products_Purchased__c,"Super Dust Tea Powder"),1,0) +
IF(INCLUDES(Products_Purchased__c,"Osmania Biscuits (Value Pack)"),1,0)
```

Return Type → **Number**

17.6. TOTAL BILL CALCULATION FORMULA

Automatically calculates **bill amount**.

```
IF(INCLUDES(Products_Purchased__c,"Platinum Tea Powder"),175,0) +
IF(INCLUDES(Products_Purchased__c,"Double Delight Tea Powder"),150,0) +
```

```
IF(INCLUDES(Products_Purchased__c,"Super Dust Tea Powder"),125,0) +
IF(INCLUDES(Products_Purchased__c,"Osmania Biscuits (Value Pack)"),50,0)
```

Return Type → **Currency**

17.7. SCHEDULED FLOW LOGIC (PSEUDO CODE)

Used for **Birthday Email Automation**

```
Start Flow Daily

IF
Customer.Date_of_Birthday_is_today = TRUE
AND
Customer.Email IS NOT NULL

THEN
Send Email Alert (Birthday Offer Email)

END
```

17.8. EMAIL TEMPLATE CODE

```
Dear {!Customer_Account__c.First_Name__c},

Happy Birthday from TheChaiShop&Co!

We are happy to offer you a special 50% birthday discount.

Use Code: HBDAY50

Thank you for being our valued customer.

Regards
TheChaiShop&Co
```

17.9. DATA FLOW LOGIC

User Login
↓
Salesforce UI
↓
Custom Objects
↓
Validation Rules
↓
Formula Calculations
↓
Flows Automation
↓
Reports
↓
Dashboards

18. AUTOMATION USING FLOWS

Automation in Salesforce is used to perform tasks automatically without manual effort. It helps save time, reduce human errors, and improve efficiency in business processes. Salesforce provides a powerful tool called Flows that allows developers and administrators to automate actions such as sending emails, updating records, or triggering notifications.

In the CRM Application for TheChaiShop&Co, automation is used to send birthday greeting emails to customers automatically. This helps improve customer relationships and engagement by sending personalized messages on special occasions.

Steps to Create Birthday Email Automation

1. Create Schedule Triggered Flow

First, a Schedule Triggered Flow is created in Salesforce.

- Go to Setup in Salesforce

- Open Flows
- Click New Flow
- Select Schedule Triggered Flow
- Set the schedule to run daily

This flow will automatically run every day and check customer records.

2. Check Birthday Condition

After scheduling the flow, a condition is added to check if the customer's birthday matches the current date.

Example condition:

- Customer Date of Birth = Today's Date

If the condition is true, the system proceeds to the next step.

3. Trigger Email Alert

Once the birthday condition is satisfied:

- The flow triggers an Email Alert
- A predefined email template is sent to the customer
- The email contains a birthday greeting message from TheChaiShop&Co

This ensures customers receive greetings automatically on their birthday.

Flow Process Summary

Step	Process	Description
1	Schedule Trigger	Flow runs automatically every day
2	Data Check	System checks customer records for birthdays

3	Condition Validation	Compares the birth date with the current date
4	Email Alert	Sends a birthday greeting email to the customer

18.1 BENEFITS OF AUTOMATION USING FLOWS

- Reduces manual work
- Improves customer engagement
- Ensures timely communication
- Enhances customer relationship management

19. REPORTS AND DASHBOARDS

Reports and Dashboards in Salesforce are used for data analysis and business insights. They help organizations understand customer behavior, sales performance, and business growth by presenting data in an organized and visual format. Reports collect and display data from objects, while dashboards present that data using charts, graphs, and visual components.

In the CRM Application for TheChaiShop&Co, reports and dashboards help managers and analysts monitor sales activities and make better business decisions.

19.1 REPORTS

A report is a structured list of records that shows data stored in the Salesforce system. Reports help users analyze information such as customer purchases, sales performance, and order history.

The following reports were created in the CRM system:

Report Name	Description
Customer Sales Report	Displays sales data related to customers, including order details, purchase amounts, and transaction history.
Enterprise Sales Report	Shows sales information related to enterprise accounts, including business transactions and revenue generated from enterprise clients.

These reports help managers track sales trends and evaluate customer and enterprise business performance.

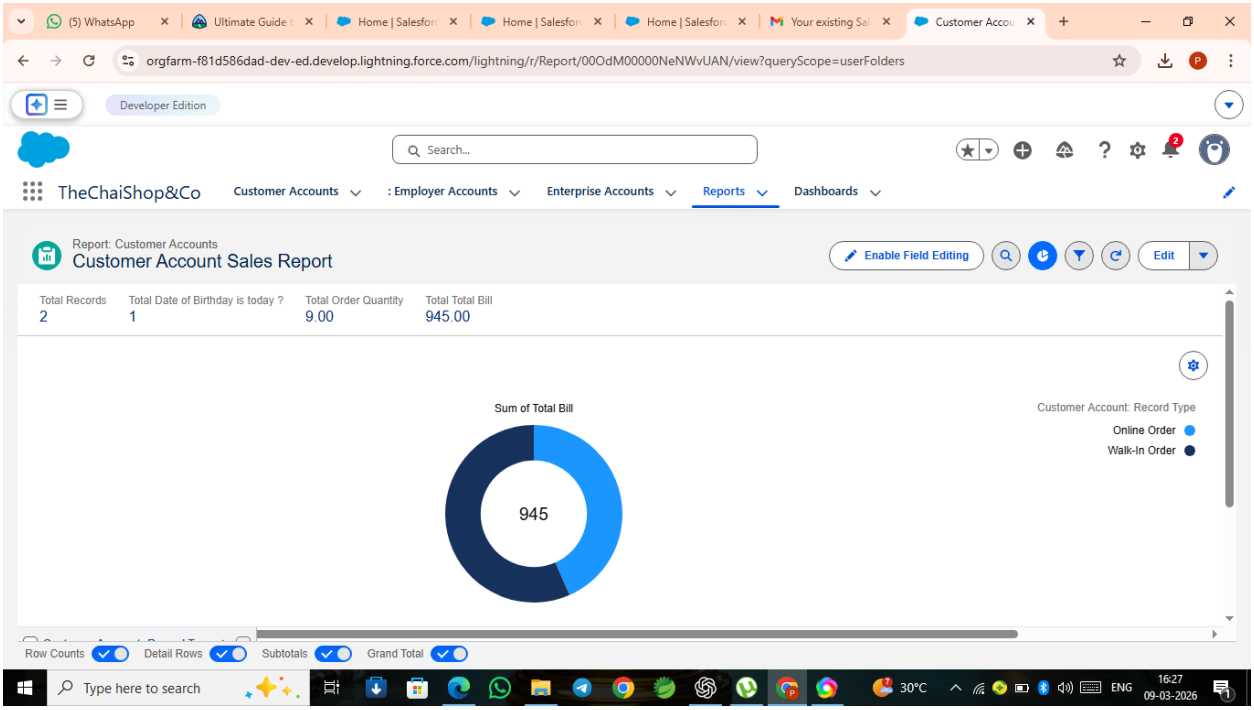


Fig: 19.1.1

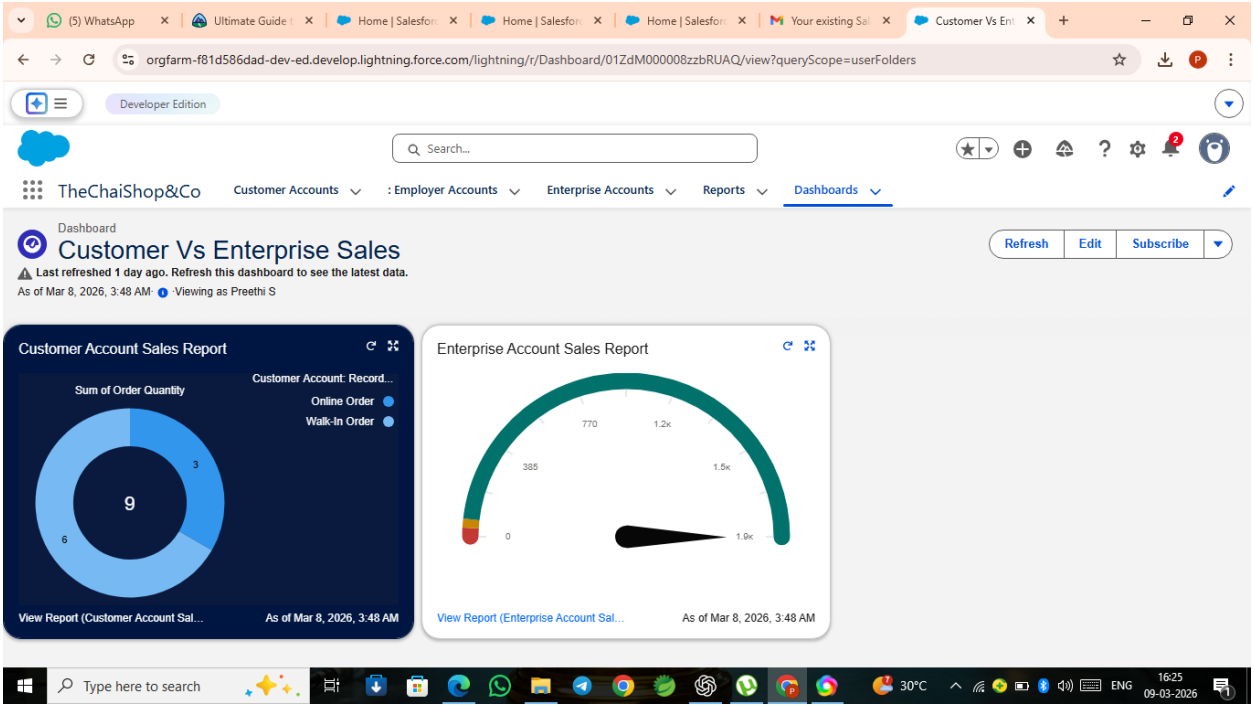
19.2 DASHBOARD

A dashboard is a visual representation of reports. It displays key information using charts, graphs, and visual indicators. Dashboards provide a quick overview of important business metrics.

In this CRM system, the following dashboard was created:

Dashboard Name	Description
Customer vs Enterprise Sales	Compares the sales performance between customer accounts and enterprise accounts using visual charts.

This dashboard helps managers quickly understand which type of account generates more sales.



19.2.1

19.3 BENEFITS OF DATA VISUALIZATION

Data visualization plays an important role in business analytics because it makes complex data easier to understand.

Benefit	Description
Easy Data Understanding	Charts and graphs make data easier to interpret compared to large tables of numbers.
Quick Decision Making	Managers can quickly identify trends and make business decisions.
Better Data Comparison	Visualization allows easy comparison between different sales categories.
Improved Business Insights	Helps identify patterns, opportunities, and areas for improvement.

20. TESTING, ADVANTAGES & FUTURE SCOPE

20.1 TESTING

Testing is an important phase in software development that ensures the system works correctly and meets the project requirements. Testing helps identify errors, verify system functionality, and improve the reliability of the CRM application. In the CRM Application for TheChaiShop&Co, different types of testing were performed to ensure smooth operation and accurate data management.

Testing Type	Description
Functional Testing	Functional testing verifies that all system features work correctly. It checks whether functions such as user login, customer data entry, order creation, billing calculation, and report generation operate as expected.
Validation Testing	Validation testing ensures that the data entered into the system follows predefined rules. For example, validation rules check whether Aadhaar numbers contain 12 digits, phone numbers have the correct format, and PAN numbers follow the correct pattern.
Performance Testing	Performance testing evaluates how well the system performs under different conditions. It checks system speed, response time, and the ability to handle multiple users simultaneously.

20.2 ADVANTAGES

The CRM Application provides several benefits that improve business operations and customer relationship management.

Advantage	Description
Automated Billing	The system automatically calculates billing amounts, reducing manual errors and improving accuracy.
Faster Data Access	Customer and sales information can be retrieved quickly from the centralized database.
Better Customer Engagement	Automated email notifications and personalized communication help strengthen customer relationships.

20.3 DISADVANTAGES

Although the CRM system provides many advantages, it also has a few limitations.

Disadvantage	Description
Internet Dependency	Since Salesforce is a cloud-based platform, a stable internet connection is required to access the system.
Salesforce License Cost	Advanced Salesforce features may require paid licenses, which can increase operational costs for businesses.

20.4 FUTURE SCOPE

The CRM application can be further improved by adding new features and technologies in the future.

Future Feature	Description
Mobile CRM	Developing a mobile version of the CRM system will allow users to access customer data and manage orders using smartphones.
AI Sales Prediction	Artificial intelligence can be integrated to analyze sales data and predict future sales trends.
WhatsApp Notifications	Integrating WhatsApp messaging can enable instant communication with customers for offers, updates, and notifications.
Inventory Management	Adding inventory management features will help track product stock levels and automatically update inventory after each sale.

21. CONCLUSION

The CRM Application for TheChaiShop&Co was developed to improve customer relationship management and streamline business operations using the Salesforce platform. The system replaces traditional manual processes with a modern digital solution that efficiently manages customer information, sales records, and communication. Throughout the project, important CRM functionalities such as customer data management, order creation, automated billing, email notifications, and reporting were successfully implemented. Custom objects, fields, validation rules, and automation using Salesforce Flows were created to ensure accurate data management and efficient system operations.

The system also includes reports and dashboards that help managers and analysts monitor sales performance and customer activity. These analytical tools support better business decision-making by providing clear insights into customer and enterprise sales data. Testing methods such as functional testing, validation testing, and performance testing were conducted to ensure the reliability and efficiency of the system. The CRM application offers several advantages, including faster data access, automated processes, and improved customer engagement.